

Final Exam
COSE215: Theory of Computation
2026 Spring

Instructor: Jihyeok Park

June 17, 2026. 13:30-14:45

- If you are not good at English, please write your answers in Korean.
(영어가 익숙하지 않은 경우, 답안을 한글로 작성해 주세요.)
- Write answers in good handwriting.
If we cannot recognize your answers, you will not get any points.
(글씨를 알아보기 힘들면 점수를 드릴 수 없습니다. 답안을 읽기 좋게 작성해주세요.)
- Write your answers in the boxes provided.
(답안을 제공된 박스 안에 작성해 주세요.)

Student ID	
Student Name	

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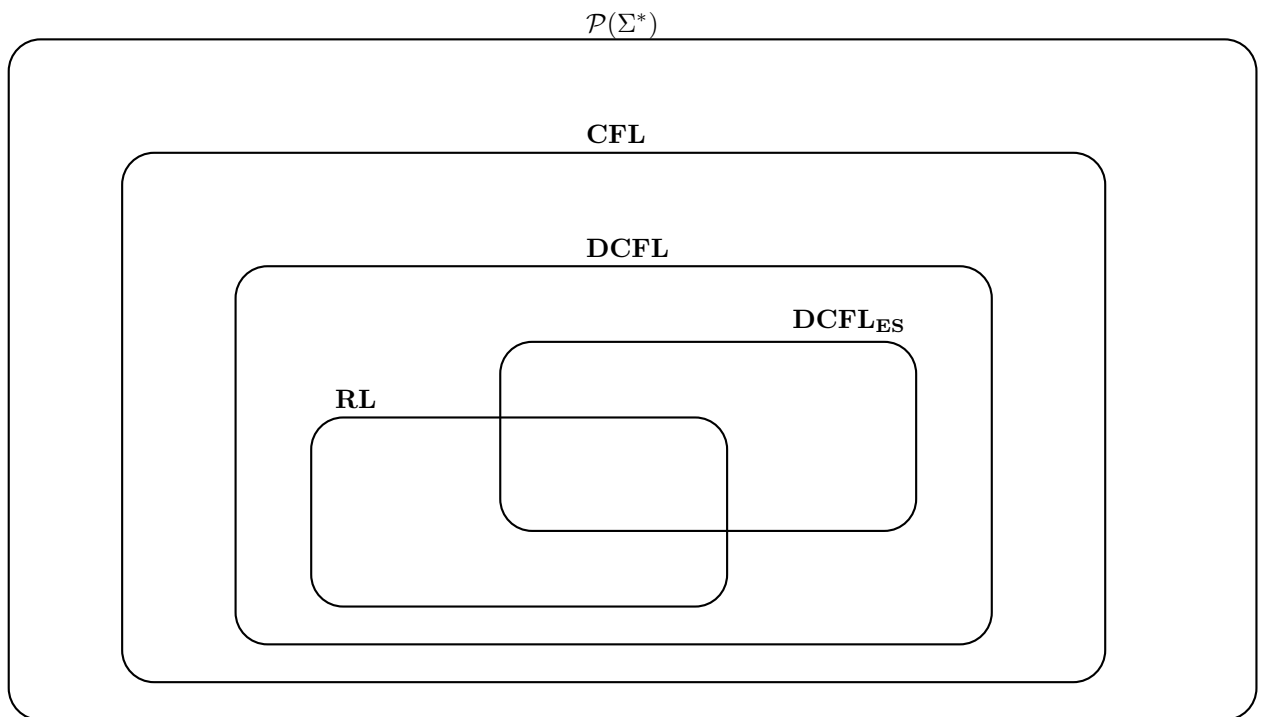
1. 15 points Consider the following languages:

- | | |
|---|--|
| 1. $L_1 = \{a^n b \mid n \geq 1\}$ | 6. $L_6 = \{a^n b^n c^n \mid n \geq 1\}$ |
| 2. $L_2 = \{ab^n \mid n \geq 1\}$ | 7. $L_7 = \{ww \mid w \in \{a, b\}^*\}$ |
| 3. $L_3 = \{a^n b^m \mid n, m \geq 0\}$ | 8. $L_8 = \{wcw^R \mid w \in \{a, b\}^*\}$ |
| 4. $L_4 = \{a^n b^{2n} \mid n \geq 1\}$ | 9. $L_9 = \{w \in \{a, b\}^* \mid w = w^R\}$ |
| 5. $L_5 = \{a^i b^j c^k \mid i + k = j\}$ | 10. $L_{10} = \{w \in \{a, b\}^* \mid w \neq xx \text{ for any } x \in \{a, b\}^*\}$ |

(a) 10 points [★☆☆] Consider the following classes of languages.

- **CFL**: the class of **context-free languages**.
- **DCFL**: the class of **deterministic context-free languages**.
- **DCFL_{ES}**: the class of **deterministic context-free languages by empty stacks**.
- **RL**: the class of **regular languages**.

The following Venn diagram shows the relationships between these language classes. Place the above languages in the following Venn diagram using their numbers (e.g., 1, 2, ...).



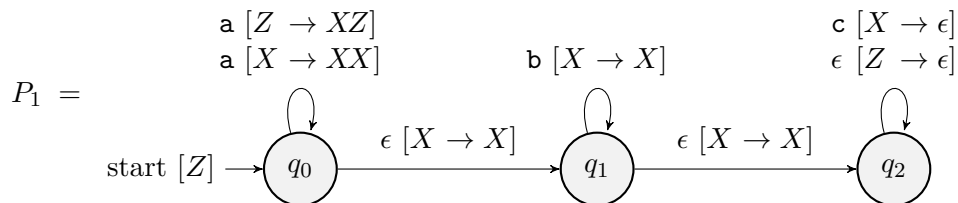
(b) 5 points [★☆☆] Some languages are **NOT DCFLs** but can still be defined by an **unambiguous CFG**. Find such a language in the **above list** and construct an **unambiguous CFG** for the language.

2. 10 points [★☆☆] **True/False questions.** Answer O for True and X for False.

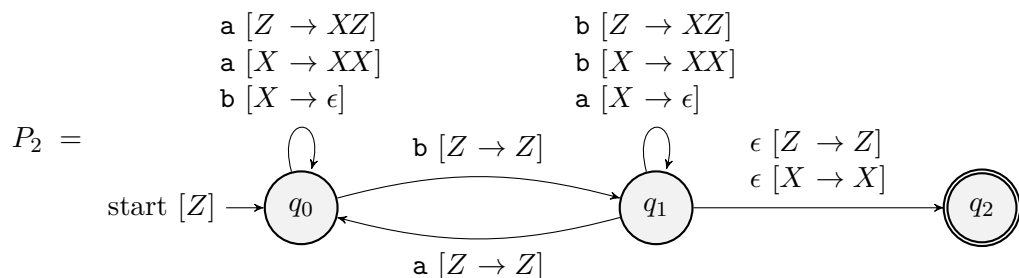
1. The class of CFLs is closed under homomorphisms. ☐
2. The class of CFLs is closed under intersection. ☐
3. The class of DCFLs is closed under complement. ☐
4. The number of PDAs is countably infinite. ☐
5. The cardinality of the set of all languages $\mathcal{P}(\Sigma^*)$ over a finite set Σ of symbols is $\aleph_0$. ☐
6. We can prove a language is context-free using pumping lemma for CFLs. ☐
7. All NP-hard problems are NP problems. ☐
8. There exists an undecidable language that is recognized by a Turing machine. ☐
9. All NP-complete problems are polynomial-time reducible to each other. ☐
10. If $P = NP$, then all NP problems are decidable in polynomial time. ☐

3. 10 points Explain the languages defined by the **empty stacks** (L_E) or **final states** (L_F) of PDAs P_i .

(a) 5 points [★☆☆] $L_E(P_1) = \left\{ \begin{array}{c} \text{ } \end{array} \right\}$



(b) 5 points [★★☆] $L_F(P_2) = \left\{ \begin{array}{c} \text{ } \end{array} \right\}$



4. 15 points We can freely **convert** a PDA to an **equivalent** CFG and vice versa.

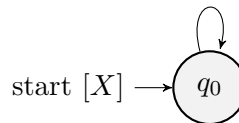
(a) 5 points [☆☆☆] Consider the following CFG:

$$S \rightarrow \epsilon \mid aSa \mid bSb$$

Construct a PDA accepting the language of the above CFG by its **empty stacks** with a **single state**.

(b) 5 points [★★☆] Consider the following PDA P that accepts a language by its **empty stack**:

$$P = \begin{array}{l} \mathbf{a} \ [X \rightarrow YXY] \\ \mathbf{b} \ [X \rightarrow XX] \\ \mathbf{c} \ [Y \rightarrow \epsilon] \\ \epsilon \ [X \rightarrow \epsilon] \\ \epsilon \ [Y \rightarrow YY] \end{array}$$



Fill in the blanks in the production rules of a CFG that represents the language accepted by the **empty stack** of the PDA, where each variable is a **stack symbol** satisfying the following:

$$X \Rightarrow^* w \quad \text{if and only if} \quad (q_0, w, X) \vdash^* (q_0, \epsilon, \epsilon)$$

$X \rightarrow$	
$Y \rightarrow$	

(c) 5 points [★★☆] In general, explain an **algorithm** that converts a PDA $P = (\{q_0\}, \Sigma, \Gamma, \delta, q_0, Z, \emptyset)$ with a **single state** accepting by **empty stack** into an equivalent CFG $G = (V, \Sigma, S, R)$.

5. 15 points Please construct new CFGs (G'_0 , G'_1 , and G'_2) from the given CFG (G_0 , G_1 , and G_2) by applying the following transformations.

- (a) 5 points [☆☆☆] Construct a CFG G'_0 consisting of productions produced by replacing **nullable variables** with ϵ in all combinations and removing all ϵ -productions in production rules in G_0 .

$$G_0 = \begin{cases} S \rightarrow AAB \\ A \rightarrow aA \mid \epsilon \\ B \rightarrow bB \mid b \end{cases}$$

$G'_0 =$

- (b) 5 points [☆☆☆] Construct a CFG G'_1 by removing all **unit productions** and adding all possible **non-unit productions** of Y to X for each **unit pair** (Y, X) in G_1 .

$$G_1 = \begin{cases} S \rightarrow A \mid aB \\ A \rightarrow bA \mid B \\ B \rightarrow b \end{cases}$$

$G'_1 =$

- (c) 5 points [★☆☆] Construct a CFG G'_2 by removing all productions that contain **non-generating variables** or come from **unreachable variables** in G_2 .

$$G_2 = \begin{cases} S \rightarrow AB \mid CC \\ A \rightarrow aA \mid a \\ B \rightarrow bB \\ C \rightarrow aCa \mid DD \\ D \rightarrow bb \\ E \rightarrow aE \end{cases}$$

$G'_2 =$

6. 10 points [★★☆] Fill in the blanks in the **proof** showing that the language L is **not** a **context-free language (CFL)** using the **pumping lemma** for **CFLs**.

$$L = \{a^p \mid p \text{ is a prime number}\}$$

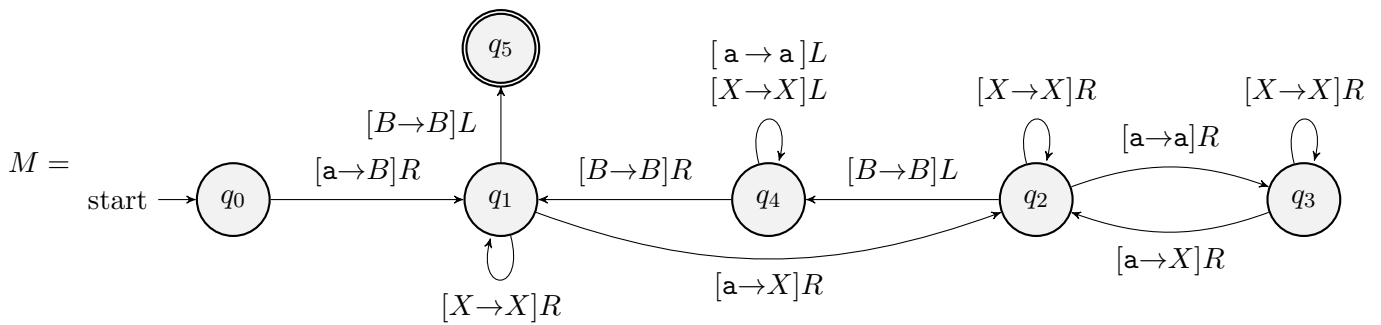
1. Assume that any positive integer n is given. (i.e., $n \geq 1$)
2. Pick a word $L \ni z =$.
3. $|z| =$ $\geq n$.
4. Assume that any split $z = uvwxy$ satisfying ① $|vx| > 0$ and ② $|vwx| \leq n$ is given.
5. We need to show that $\neg$ ③ $uv^iwx^iy \notin L$ for some $i \geq 0$:

7. 10 points [★★☆] Draw a transition diagram of a **Turing machine** that **computes** the following function:

$$f : \{\#1^n \mid n \geq 0\} \rightarrow \{0, 1, \#\}^* \text{ s.t. } f(\#1^n) = (\text{the binary representation of } n)$$

For example, $f(\#) = \epsilon$, $f(\#1) = 1$, $f(\#11) = 10$, $f(\#111) = 11$, $f(\#1^5) = 101$, and $f(\#1^{13}) = 1101$.

8. 8 points [★★★] Write the **time complexity** of the following **Turing machine** M in a big-O notation in terms of $n = |w|$ for any input $w \in \{a\}^*$ and **explain** the reasoning behind your answer.



Time complexity of $M =$

9. 7 points [★★★] **Prove** or **disprove** the following statement:

The **recursively enumerable language (REL)** is closed under **complement**.

(Hint: The **universal language** $L_u = \{(M, w) \mid w \in L(M) \text{ where } M \text{ is a TM and } w \text{ is a binary word}\}$)

This is the last page.
I hope that your tests went well!